

# Manual

## Statistical Process Analysis PDV-SPA 8.3

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# 1 Overview: Statistical Process Analysis

## Overview

### Purpose

This product includes the statistical process data logs. You use the applications to analyze the collected process data of orders and operations. The data is displayed in a tabular overview that shows the statistical parameters of the process.

### Integration

For this product, the following requirements must be fulfilled:

- machine interface for data collection in the Process Communication Controller (PCC),
- license PDV-PDM to collect and process process data.

### Features

This product provides evaluation options of the process data logs:

- Process data log for order: evaluation of the order-related statistical parameters using the collected process data of an order.
- Process data log for operation: evaluation of the operation-related statistical parameters using the collected process data of an operation.

## 2 Process Data Log for Operation

### Overview

|                        |                                                                        |
|------------------------|------------------------------------------------------------------------|
| Menu                   | Quality management → Process analysis → Process data log for operation |
| Transaction code       | pdsop                                                                  |
| Function authorization | pdsop                                                                  |

The application *Process data log for operation* provides a quick overview of the process data recorded for an operation.

### Purpose

The application provides an analysis of the process data recorded for an operation. The data is displayed in an operation-specific tabular overview.

### Integration

The function shows the process parameters recorded in combination with the data of the shop floor data collection.

### Requirements

To use the application, the process and the shop floor data (orders/operations) must be collected.

### Selection criteria

The application provides the following selection criteria:

#### Operation

This selection criterion defines the operation for which the measured values are displayed. The operation is specified via MES order number (combined order and operation number).

#### Period from / until

This selection criterion defines the evaluation period for which the measured values are displayed.

#### Consider long-term data

If you activate this selection parameter, the long-term data is also used in the data selection to identify the article number and name.



To identify the operations of the period selected, only the operation logon time is used. The operation logoff time is not relevant for the selection.

## Field descriptions

### Operation

The operation is specified via MES order number (combined order and operation number).

### OP

This field shows the operation number for which the process parameter was recorded.

### Article number of OP

This field provides the article number of the specified operation. The *Article number of OP* is specified via the data of the backlog of orders.

### Article designation/name

This field provides the article name of the specified operation. The *Article name* is specified via the data of the backlog of orders.

### Process parameter

This field specifies the process parameter recorded for the specified order.

### Mean value (AVG - order sequencing list)

This field shows all calculated mean values of all measured values recorded for the specified process parameter.

### Maximum value

This field shows the maximum value of all measured values recorded for the specified process parameter.

### Minimum value

This field shows the minimum value of all measured values recorded for the specified process parameter.

### Range of values

The value range is calculated using the difference between maximum and minimum value.

### Number of measured values

This field shows the number of measured values recorded for the specified process parameter.

### Standard deviation

This field shows the calculated standard deviation of the measured values of the relevant process parameter.

**cp**

The process capability index  $C_p$  is shown in the column Cp. The index for process capability cp is calculated using the following factors:

- Upper and lower tolerance limit
- Standard deviation.

**cpk**

The process capability index  $C_{pK}$  is shown in the column cpk. The index for process capability cpk is calculated using the following factors:

- Mean value
- Standard deviation
- Upper and lower tolerance limit

A high cpk value shows that the production lies solidly within the specification limits.

**Lower tolerance limit**

This field shows the value of the lower tolerance limit for the specified process parameter. The value that was valid at the last operation logoff is used.

**Lower process action limit**

This field shows the value of the lower process action limit for the specified process parameter. The value that was valid at the last operation logoff is used.

**Target value**

This field shows the target value of the specified process parameter. The value that was valid at the last operation logoff is used.

**Upper process action limit**

This field shows the value of the upper process action limit for the specified process parameter. The value that was valid at the last operation logoff is used.

**Upper tolerance limit**

This field shows the value of the upper tolerance limit for the specified process parameter. The value that was valid at the last operation logoff is used.

**Distribution**

Graphic display of the distribution of the measured values for the specified process parameter.



The fields *Article number of OP* and *Article name* are selected from the data of the order backlog. If this data is archived and not available in the HYDRA database, then these fields remain empty.



## 3 Process Data Log for Order

### Overview

|                        |                                                                    |
|------------------------|--------------------------------------------------------------------|
| Menu                   | Quality management → Process analysis → Process data log for order |
| Transaction code       | pdsor                                                              |
| Function authorization | pdsor                                                              |

The application *Process data log for order* provides a quick overview of the process data recorded for an order.

### Purpose

The application provides an analysis of the process data recorded for an order. The data is displayed in an order-specific tabular overview. The data is displayed in combination with the statistical parameters of the process parameter.

### Integration

The function shows the process parameters recorded in combination with the data of the shop floor data collection.

### Requirements

To use the application, the process and the shop floor data (orders/operations) must be collected.

### Selection criteria

The application provides the following selection criteria:

#### Order

This selection criterion defines the order for which the measured values are displayed.

#### Period from / until

This selection criterion defines the evaluation period for which the measured values are displayed.

#### Consider long-term data

If you activate this selection parameter, the long-term data is also used in the data selection to identify the article number and name.

## Field descriptions

### Order

This field shows the order number for which the process parameter has been collected.

### Final article

This field shows the final article of the specified order. The *Final article* is specified via the data of the backlog of orders.

### Article designation/name

This field shows the article name of the specified order. The *Article name* is specified via the data of the backlog of orders.

### Process parameter

This field specifies the process parameter recorded for the specified order.

### Mean value

This field shows all calculated mean values of all measured values recorded for the specified process parameter.

### Maximum value

This field shows the maximum value of all measured values recorded for the specified process parameter.

### Minimum value

This field shows the minimum value of all measured values recorded for the specified process parameter.

### Range of values

The value range is calculated using the difference between maximum and minimum value.

### Quantity

This field shows the number of measured values recorded for the specified process parameter.

### Standard deviation

This field shows the calculated standard deviation of the measured values of the relevant process parameter.

### cp

The process capability index Cp is shown in the column Cp. The index for process capability cp is calculated using the following factors:

- Upper and lower tolerance limit
- Standard deviation.

**cpk**

The process capability index  $CpK$  is shown in the column cpk. The index for process capability cpk is calculated using the following factors:

- Mean value
- Standard deviation
- Upper and lower tolerance limit

A high cpk value shows that the production lies solidly within the specification limits.

**Lower tolerance limit**

This field shows the value of the lower tolerance limit for the specified process parameter. The system uses the value that was valid at the last order logoff.

**Lower process action limit**

This field shows the value of the lower process action limit for the specified process parameter. The system uses the value that was valid at the last order logoff.

**Target value**

This field shows the target value of the specified process parameter. The system uses the value that was valid at the last order logoff.

**Upper process action limit**

This field shows the value of the upper process action limit for the specified process parameter. The system uses the value that was valid at the last order logoff.

**Upper tolerance limit**

This field shows the value of the upper tolerance limit for the specified process parameter. The system uses the value that was valid at the last order logoff.

**Distribution**

Graphic display of the distribution of the measured values for the specified process parameter.



The fields *Final article* and *Article name* are selected from the data of the order backlog. If this data is archived and not available in the HYDRA database, then these fields remain empty.